Simple and Multiple Regression Models: Estimation

These notes accompany Lecture 2. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

Learning outcomes – after this lecture you should be able to:

1. Write down the simple and multiple regression models and name every component.
2. Explain what the OLS estimator does, and derive the interpretation of each coefficient.
3. Interpret a slope coefficient as a ceteris paribus (partial) effect.
4. State the five Gauss-Markov assumptions and say what breaks when each one fails.
5. Determine the direction of omitted variable bias.
6. Predict how changing units of measurement affects the estimates, and interpret level-level, log-level, level-log and log-log models.

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1. The simple regression model

1.1 Why economists are obsessed with regressions

Countries that trade more with the rest of the world also have higher income levels – but does this mean that trade raises income? People with more education tend to have higher earnings – but does that imply education causes higher earnings?

To answer questions of this type, economists use regression analysis. Regressions quantify the relationship between one variable and the other variables thought to explain it, and they also tell us how close and how well determined that relationship is.

> **The role of regression**
>
> Regression is the basic statistical tool for distinguishing between correlation and causality – the problem introduced in Lecture 1.

1.2 Where the word "regression" comes from

The name is a historical accident, and knowing its origin tells you something real about the method.

Francis Galton (1822–1911), a cousin of Charles Darwin, spent the 1870s and 1880s studying heredity. He measured sweet pea seeds, dogs, and eventually about a thousand human families – parents' heights against their adult children's heights. He expected tall parents to produce tall children, and they did. But he noticed something he had not expected: the children of very tall parents were on average shorter than their parents, and the children of very short parents were on average taller than theirs. Extreme values in one generation pulled back towards the population average in the next.

Galton called this "regression toward mediocrity" – mediocrity in the old sense, meaning simply the mean. That is literally where our term comes from.

Galton first read this as a biological force pulling traits back towards a species type. It is not. It is a purely statistical phenomenon: whenever two variables are imperfectly correlated, extreme values on one will on average be associated with less extreme values on the other. Nothing biological is required.

> **A common misreading**
>
> Regression to the mean does not imply that human heights are converging over generations. The population's spread stays stable, because average parents also produce some extreme children.

1.3 Definition of the simple regression model

Suppose y and x are two variables representing some population, and we want to explain y in terms of x, or study how y varies with changes in x. The equation below, assumed to hold in the population of interest, defines the **simple linear regression model**:

$$y = \beta_0 + \beta_1 x + u$$

Component	Names	Meaning
y	Dependent, explained variable, regressand	The variable we are trying to explain
x	Independent, explanatory variable, regressor	The variable doing the explaining
u	Error term, disturbance	All unobserved factors affecting y
β_0	Intercept, constant term	Value of y when x = 0
β_1	Slope parameter	The ceteris paribus effect of x on y

As long as the intercept β_0 is included, nothing is lost by assuming the average value of *u* in the population is zero: E(*u*) = 0.

The slope is the parameter of primary interest. If the other factors in u are held fixed, so that $\Delta u = 0$, then x has a linear effect on y:

$$\Delta y = \beta_1 \Delta x \quad (\text{if } \Delta u = 0)$$

The intercept β_0 shows the value of y when x = 0. In most applications it has no useful economic interpretation, but it is needed to construct the best-fitting regression line.

1.4 Obtaining the OLS estimates

The population regression equation is unknown and will remain unknown. What we can do is estimate it from a sample. Because of sampling fluctuation, we cannot expect to recover it exactly.

Take a random sample $\{(x_i, y_i): i = 1, \ldots, n\}$ from the population. For each observation:

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

Here u_i is the error term for observation *i*, containing everything other than x_i that affects y_i.

The estimated version of the population regression function is called the **sample regression function** (SRF):

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

> **Keep the two levels apart**
>
> Population regression function (PRF): $y = \beta_0 + \beta_1 x + u$ – unknown, contains the true parameters and the error term u.
>
> Sample regression function (SRF): $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ – computed from data, contains estimates and produces residuals $\hat{u}$.
>
> The parameters β are fixed and unknown; the estimates $\hat{\beta}$ are random variables that change from sample to sample.

1.5 The least squares principle – and where it came from

On 1 January 1801 the Italian astronomer Giuseppe Piazzi spotted something moving against the fixed stars. Astronomers had been hunting for a "missing planet" between Mars and Jupiter, and this looked like it. He named it Ceres and tracked it for forty-one nights – then it drifted too close to the Sun and disappeared. When the sky cleared months later, nobody could find it again.

Piazzi had watched Ceres cross only about 3 degrees of sky, roughly one percent of its orbit. From that sliver, astronomers had to predict where the object would reappear months later. Europe's best tried, and failed.

Carl Friedrich Gauss, then twenty-four, had a method nobody else had – one he had been sitting on since he was about eighteen. The problem he faced is exactly our problem: Piazzi's observations were more numerous than the unknowns in the orbit, and they contradicted each other, because each carried small measurement errors. No orbit fitted all the data exactly.

Most astronomers responded by throwing out observations, keeping just enough to solve the equations cleanly. Gauss did the opposite. He kept everything, accepted that no curve would pass through every point, and asked a different question: which orbit makes the sum of squared errors as small as possible? He computed for weeks and sent a prediction. On the last night of 1801, Franz Xaver von Zach pointed his telescope where Gauss had said to look – and Ceres was there.

> **Why the story matters**
>
> The method mattered more than the planet. It is the same method we use in this course: you have noisy data, more observations than parameters, and no perfect fit – so you choose the line that minimises the sum of squared residuals.

The OLS criterion

The method of **ordinary least squares (OLS)** chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ simultaneously to minimise the sum of squared residuals:

$$\Sigma\, \hat{u}_i^2 \;=\; \Sigma\, (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

where the **residual** for observation *i* is $\hat{u}_i = y_i - \hat{y}_i$.

Note carefully: the residual $\hat{u}$ is not the error term u. The error term is unobservable and belongs to the population model; the residual is computed from the fitted line and is observable.

1.6 The estimators and their interpretation

Minimising the sum of squared residuals gives:

$$\hat{\beta}_1 = \Sigma\, (x_i - \bar{x})(y_i - \bar{y}) \;/\; \Sigma\, (x_i - \bar{x})^2$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{y}$ is the sample average of the y_i and likewise for $\bar{x}$.

The slope estimate has a **partial effect** (or ***ceteris paribus***) interpretation: $\hat{\beta}_1$ measures the change in $\hat{y}$ due to a one-unit increase in x, holding all other factors fixed:

$$\Delta\hat{y} = \hat{\beta}_1 \Delta x$$

1.7 Example: wage and education

Let y = wage, the wage in dollars per hour, and x = educ, years of schooling, for a sample of n = 526 workers. OLS gives the regression line:

$$\widehat{wage} = -0.90 + 0.54\,educ$$

Interpret this with caution:

- The intercept of −0.90 says a person with no education has a predicted hourly wage of −90 cents. This is silly – it reflects the fact that the model is being extrapolated far outside the range of the data.
- The slope says one more year of education increases the predicted hourly wage by 54 cents. Four more years therefore raise the predicted wage by 4 × 0.54 = \$2.16 per hour. These are fairly large effects.

Notice a limitation built into the linear form: 54 cents is the predicted increase for the first year of education and equally for the twentieth. We return to this in Section 4.

2. The multiple regression model

2.1 Motivation

In practice it is not realistic to think we can explain the variation in an economic outcome using a single explanatory variable. Adding further factors that help explain y lets us account for more of its variation – and, more importantly, lets us hold those factors fixed.

Multiple regression analysis makes it possible to control explicitly for factors that simultaneously affect the dependent variable. Returning to the wage example:

$$wage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 tenure + u$$

where educ is years of education, exper is years of labour market experience, and tenure is years with the current employer.

> **What the extra regressors buy you**
>
> Compared with the simple regression of wage on education, this equation takes experience and tenure out of the error term and puts them explicitly in the model.
>
> We are still primarily interested in β_1 – but now it measures the effect of education holding experience and tenure fixed. The coefficient β_2 simultaneously gives the ceteris paribus effect of experience, which is of interest in its own right.

The general multiple linear regression model in the population is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u$$

With k independent variables and an intercept, the model contains $k + 1$ unknown population parameters. As in simple regression, u is the error term, containing all factors other than $x_1, \ldots, x_k$ that affect y.

2.2 Obtaining the OLS estimates

The sample regression function is:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_k x_k$$

The k + 1 OLS estimates are chosen, exactly as before, to minimise the sum of squared residuals:

$$\Sigma\, \hat{u}_i^2 = \Sigma\, (y_i - \hat{y}_i)^2 = \Sigma\, (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \ldots - \hat{\beta}_k x_{ik})^2$$

2.3 Interpreting the estimated equation

The coefficient on x_1 measures the change in $\hat{y}$ due to a one-unit increase in x_1, holding x_2, x_3, …, x_k fixed:

$$\Delta\hat{y} = \hat{\beta}_1 \Delta x_1$$

We say we have controlled for the other variables when estimating this effect. The remaining coefficients have the same interpretation – each is a partial effect.

The intercept $\hat{\beta}_0$ is the predicted value of y when all the x's equal zero. Sometimes that is an interesting scenario; often it is not. Either way, the intercept is needed to obtain predictions from the line.

Example

Using the same 526 observations on workers, and including education, experience and tenure:

$$\textbf{wage} = -2.87 + 0.60\ \textbf{educ} + 0.02\ \textbf{exper} + 0.17\ \textbf{tenure}$$

The coefficient 0.60 means that, holding experience and tenure fixed, another year of education is predicted to increase the wage by 60 cents per hour.

> **Compare with the simple regression**
>
> Simple regression: wage = −0.90 + 0.54 educ
>
> Multiple regression: wage = −2.87 + 0.60 educ + 0.02 exper + 0.17 tenure
>
> The education coefficient changes from 0.54 to 0.60 once experience and tenure are controlled for. Section 3.6 explains why coefficients move like this.

2.4 Fitted values and residuals

For observation i the fitted (predicted) value is:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \ldots + \hat{\beta}_k x_{ik}$$

and the residual is:

$$\hat{u}_i = y_i - \hat{y}_i$$

Normally the actual value will not equal the predicted value. OLS minimises the average squared prediction error, which says nothing about the error for any particular observation. Do not forget the intercept when computing fitted values, or the answer can be badly misleading.

A useful algebraic fact: the sample average of the OLS residuals is zero (provided an intercept is included).

2.5 Goodness of fit

We can split the total variation in y into an explained part and an unexplained part. With SST = SSE + SSR:

Symbol	Name	Definition
SST	Total sum of squares	$\Sigma (y_i - \bar{y})^2$
SSE	Explained sum of squares	$\Sigma (\hat{y}_i - \bar{y})^2$
SSR	Residual sum of squares	$\Sigma \hat{u}_i^2$

The **R-squared** (or coefficient of determination) is:

$$R^2 = SSE / SST = 1 - SSR / SST$$

It measures the fraction of the sample variation in y that is explained by the model. When an intercept is included, $0 \leq R^2 \leq 1$. A value near zero indicates a poor fit of the OLS line.

Three cautions about R^2

- **R^2 never decreases when a variable is added.** It usually increases. This follows algebraically, because the sum of squared residuals never rises when extra regressors are added. R^2 is therefore a poor tool for deciding whether a variable belongs in the model.
- **What should decide inclusion** is whether the explanatory variable has a nonzero partial effect on y in the population – a question about the parameter, not about fit.
- **A low R^2 is not fatal.** Low R^2 values are common in the social sciences, especially in cross-sections. Even when the regressors collectively explain little of the variation in y, the OLS estimates can still be reliable estimates of the partial effects.

Adjusted R-squared corrects for the number of regressors (that is, adjusts for degrees of freedom) and so does not automatically rise when a variable is added. We return to model selection later in the course.

3. Assumptions of the classical linear regression model

We now turn to the statistical properties of OLS. We state five assumptions; the first four deliver unbiasedness, and the fifth adds efficiency.

> **An important warning about what "unbiased" means**
>
> Statistical properties have nothing to do with any particular sample. They are properties of the estimator across repeated random sampling.
>
> It is not meaningful to talk about the statistical properties of a set of estimates obtained from a single sample.

MLR.1 Linear in parameters

The population model can be written as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k + u$$

The key feature is that the model is linear in the parameters. It need not be linear in the variables: OLS is flexible, because y and the regressors may be arbitrary functions of the underlying variables – logarithms, squares, and so on.

> **Linear in parameters, not in variables**
>
> OLS cannot estimate a model containing a term like $\beta^3 x$, because that is nonlinear in the parameter β.
>
> OLS can estimate βx^3 or $\beta \cdot \ln(x)$, because these are linear in β even though they are nonlinear in x.

MLR.2 Random sampling

We have a random sample of n observations $\{(x_{i1}, x_{i2}, \ldots, x_{ik}, y_i): i = 1, 2, \ldots, n\}$ following the population model in MLR.1. Not all cross-sectional samples can be viewed as random samples, but many can.

Random sampling is what makes the sample representative of the population, so that the estimates computed from it carry information about the population parameters rather than about the peculiarities of whoever happened to be sampled.

MLR.3 No perfect collinearity

In the sample (and therefore in the population), none of the independent variables is constant, and there are no exact linear relationships among the independent variables.

If an independent variable is an exact linear combination of the others, the model suffers from perfect collinearity and cannot be estimated by OLS.

> **MLR.3 permits correlation – it forbids only perfect correlation**
>
> The assumption does allow the independent variables to be correlated; they just cannot be perfectly correlated. If we ruled out all correlation among regressors, multiple regression would be of very limited use.

An example of a violation

Suppose we want the effect of campaign spending on election outcomes, where voteA is the percentage of the vote for Candidate A, expendA and expendB are the two candidates' campaign expenditures, and totexpend is total campaign expenditure. A model including all three spending variables violates MLR.3, because by definition:

$$totexpend = expendA + expendB$$

If expendB and totexpend are held fixed, expendA cannot change – the ceteris paribus experiment the coefficient is supposed to describe is impossible. The solution is simple: drop any one of the three variables.

MLR.4 Zero conditional mean

The error u has an expected value of zero given any values of the independent variables:

$$E(u \mid x_1, x_2, \ldots, x_k) = 0$$

This is the final and most important assumption for unbiasedness. It is also the one that cannot be verified directly from the data, because u is unobservable – which is why defending it requires an economic argument rather than a statistical test.

Two common ways MLR.4 fails:

- **Misspecified functional form.** If the true model has log(wage) as the dependent variable but we use wage, the estimators will be biased.
- **Omitted variables.** Omitting an important factor that is correlated with any of the regressors causes MLR.4 to fail. There are always factors we cannot include, through data limitations or ignorance; the danger arises when they are correlated with the regressors.

3.5 Unbiasedness of OLS

Under assumptions MLR.1 to MLR.4, the OLS estimators are unbiased estimators of the population parameters:

$$E(\hat{\beta}_j) = \beta_j, \quad j = 0, 1, \ldots, k$$

This does not mean that the estimates from any single sample are correct. It means the procedure is unbiased when viewed across all possible random samples. We hope our sample gives an estimate close to the population value, but this cannot be assured. What is assured is that we have no reason to believe our estimate is more likely to be too big than too small.

3.6 Omitted variable bias

Suppose the true population model is:

$$wage = \beta_0 + \beta_1 educ + \beta_2 ability + u$$

Ability is not observed, so instead we estimate the simple regression:

$$\widetilde{wage} = \tilde{\beta}_0 + \tilde{\beta}_1 educ$$

The two slope estimators are related by:

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1$$

where $\hat{\beta}_1$ and $\hat{\beta}_2$ are the slope estimators from the multiple regression (if we could run it), and $\tilde{\delta}_1$ is the slope from the simple regression of ability on education. Since the full model satisfies MLR.1–MLR.4, the bias is:

$$Bias(\tilde{\beta}_1) = E(\tilde{\beta}_1) - \beta_1 = \beta_2 \tilde{\delta}_1$$

Signing the bias

The direction of the bias is the product of two signs:

	$\delta_1 > 0$ (omitted variable positively correlated with x_1)	$\delta_1 < 0$ (negatively correlated)
$\beta_2 > 0$	Positive bias – estimate too large	Negative bias – estimate too small
$\beta_2 < 0$	Negative bias – estimate too small	Positive bias – estimate too large

In the wage equation, more ability leads to higher productivity and therefore higher wages, so $\beta_2 > 0$. There are also good reasons to think education and ability are positively correlated, so $\delta_1 > 0$. Both signs are positive, so the OLS estimate of the return to education from the simple regression is on average too large.

MLR.5 Homoskedasticity

Unbiasedness is a property of central tendency. We also want to know about the spread of the estimator.

MLR.5 requires that the error has the same variance given any values of the explanatory variables:

$$Var(u \mid x_1, x_2, \ldots, x_k) = \sigma^2$$

> **Why spread matters – a question to think about**
>
> Which forecast is more useful: (i) GDP growth of 2 to 4%, or (ii) GDP growth of 1 to 5%?
>
> Both may be unbiased. The first is more precise, and precision is what makes an estimate usable.

If this assumption fails, the model exhibits heteroskedasticity. That does not bias the slope and intercept estimators – but estimators with smaller variance exist, and the usual confidence intervals and tests become unreliable.

Under MLR.1 through MLR.5, conditional on the sample values of the regressors, the sampling variance of each OLS slope estimator is:

$$Var(\hat{\beta}_j) = \sigma^2 / [\, SST_j(1 - R_j^2) \,]$$

for $j = 1, 2, \ldots, k$, where $SST_j = \Sigma(x_{ij} - \bar{x}_j)^2$ is the total sample variation in x_j, and R_j^2 is the R-squared from regressing x_j on all the other independent variables (including an intercept).

Read the formula as three separate levers on precision:

- A larger error variance σ^2 makes the estimator less precise – more noise in y that the model cannot explain.
- More variation in x_j (larger SST_j) makes it more precise – you learn more about a slope from regressors that actually move.
- Higher R_j^2 makes it less precise. As R_j^2 approaches one, the variance grows without bound. This is multicollinearity: when x_j is nearly a linear combination of the other regressors, there is little independent variation left to identify its separate effect.

3.8 The Gauss-Markov theorem

Assumptions MLR.1 to MLR.5 are together known as the Gauss-Markov assumptions (for cross-sectional regression).

There may be many unbiased estimators satisfying MLR.1 to MLR.4. Adding MLR.5 pins down which one to prefer:

Gauss-Markov theorem

Under MLR.1 through MLR.5, OLS is the Best Linear Unbiased Estimator (BLUE).

"Best" means having the smallest variance. Given two unbiased estimators, it is logical to prefer the one with the smaller variance.

Keep the consequences of the two key failures distinct:

- **Failure of MLR.4** (zero conditional mean) causes OLS to be **biased**. This is a problem about the location of the estimator, and no sample size fixes it.
- **Failure of MLR.5** (homoskedasticity) does **not** bias OLS. But OLS is no longer the minimum-variance estimator among linear unbiased estimators, and the standard errors are wrong.

3.9 Regression through the origin

Occasionally theory or common sense suggests the intercept should be zero, so we estimate:

$$\tilde{y} = \tilde{\beta}_1 x_1 + \tilde{\beta}_2 x_2 + \ldots + \tilde{\beta}_k x_k$$

where the tilde distinguishes these from the usual OLS estimates. Be warned that the familiar properties no longer hold:

- The OLS residuals no longer have a zero sample average.
- R^2 can be negative – meaning the sample average $\bar{y}$ "explains" more of the variation in y than the explanatory variables do. If that happens, either include an intercept or conclude that the regressors explain y poorly.
- If the true population intercept is different from zero, the slope estimators through the origin are biased.

By contrast, the cost of estimating an intercept when β_0 is truly zero is only that the variances of the slope estimators are somewhat larger. Since the risk is asymmetric – bias on one side, mild imprecision on the other – the default should be to include an intercept.

4. Units of measurement and functional form

Two practical issues remain: how changing the units of measurement affects the estimates, and how to build common non-linear relationships into a linear regression.

4.1 Changing the units of measurement

Regress annual CEO salary (in thousands of dollars) on return on equity (in percent):

$$\textbf{salary} = 963.19 + 18.50\ \textbf{roe}$$

The intercept 963.19 is the predicted salary when roe = 0. The slope says that if roe rises by 1 percentage point, salary rises by $18.5 thousand.

Now change the units:

Change	Resulting equation	What happened
Baseline: salary in $000s, roe in %	salary = 963.19 + 18.50 roe	–
Salary in dollars (× 1000)	salary = 963 191.3 + 18 501.2 roe	Intercept and slope both multiplied by 1000
roe as a decimal (÷ 100)	salary = 963.19 + 1850.12 (roe/100)	Slope multiplied by 100; intercept unchanged

The two general rules:

- **If the dependent variable is multiplied by a constant c,** the intercept and all slope estimates are multiplied by c.
- **If an independent variable is multiplied (divided) by a constant c,** its slope coefficient is divided (multiplied) by c. The intercept is unchanged, because roe/100 = 0 still corresponds to a zero return on equity.

The economic content of the equation never changes – only the numbers in which it is expressed. This is a useful check on your own work: if a rescaling changes a predicted value, you have made an arithmetic error.

4.2 Logarithm of the dependent variable

Recall the wage-education regression in levels:

$$\text{wage} = -0.90 + 0.54\ \text{educ}$$

Because the relationship is linear, 54 cents is the predicted increase for the first year of education and equally for the twentieth. A better description is probably that each year of education raises the wage by a constant percentage. Taking logs of the dependent variable delivers exactly that:

$$\mathbf{\log(wage) = 0.58 + 0.083\ educ}$$

Multiply the slope by 100 to read it as a percentage:

$$\%\Delta wage \approx (100 \cdot \beta_1)\,\Delta educ = 8.3\%\ \text{per year of education}$$

Wage increases by 8.3% for every additional year of education.

> **A subtlety worth noticing**
>
> Because the percentage change is constant, the dollar change is not. A constant percentage return means the absolute dollar return grows as education increases – which is the opposite of what the linear model assumed.

4.3 Constant elasticity model

Putting both variables in logs gives a constant elasticity model. Relating CEO salary to annual firm sales (in millions of dollars):

$$\mathbf{\log(salary) = 4.82 + 0.257\ \log(sales)}$$

The slope is now the elasticity of salary with respect to sales: a 1% increase in firm sales is associated with about a 0.257% increase in CEO salary – the usual interpretation of an elasticity. Compare the level-level version, salary = 1174.0 + 0.015 sales, whose slope has no such scale-free reading.

4.4 Summary of functional forms involving logarithms

Model	Dependent variable	Independent variable	Interpretation of β_1
Level-level	y	x	$\Delta y = \beta_1 \Delta x$
Level-log	y	log(x)	$\Delta y = (\beta_1/100)\ \%\Delta x$
Log-level	log(y)	x	$\%\Delta y = (100\beta_1)\ \Delta x$
Log-log	log(y)	log(x)	$\%\Delta y = \beta_1\ \%\Delta x$

In the log-level model, $100 \cdot \beta_1$ is called the semi-elasticity of y with respect to x. In the log-log model, β_1 is the elasticity. The level-log model arises less often in practice.

5. Summary

1. The multiple regression model makes it possible to hold other factors fixed while examining the effect of a particular independent variable. It explicitly allows the independent variables to be correlated with one another.
2. Although the model is linear in its parameters, it can represent non-linear relationships through an appropriate choice of the dependent and independent variables (logs, squares, and so on).
3. OLS is easily applied to estimate the model. Each slope estimate measures the partial effect of the corresponding independent variable, holding all others fixed.
4. R^2 is the proportion of the sample variation in the dependent variable explained by the independent variables. It is a goodness-of-fit measure, and it is important not to put too much weight on it when evaluating models.
5. Under MLR.1 through MLR.4 the OLS estimators are unbiased. Including an irrelevant variable does not disturb unbiasedness; omitting a relevant one causes bias, and in many cases the direction of that bias can be determined.
6. Adding an irrelevant variable generally increases the variances of the remaining OLS estimators, because of multicollinearity.
7. Under all five Gauss-Markov assumptions (MLR.1 through MLR.5), the OLS estimators are the best linear unbiased estimators (BLUE).

> **The trade-off in one line**
>
> Omitting a relevant variable risks bias. Including an irrelevant one costs precision.
>
> Bias cannot be fixed by collecting more data; imprecision can. That asymmetry should guide your choices.

Key terms

Term	Meaning
Population regression function (PRF)	$y = \beta_0 + \beta_1 x + u$. Holds in the population; the parameters are fixed and unknown.
Sample regression function (SRF)	$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$. Computed from a sample; the estimates vary from sample to sample.
Error term u	Unobservable. All factors other than the regressors that affect y.
Residual $\hat{u}$	Observable. The difference between the actual and fitted value, $y_i - \hat{y}_i$.
Ordinary least squares (OLS)	The estimator that chooses coefficients to minimise the sum of squared residuals.
Partial effect	The change in $\hat{y}$ from a one-unit change in one regressor, holding the others fixed.
SST, SSE, SSR	Total, explained and residual sums of squares. SST = SSE + SSR.
R-squared	SSE/SST. The fraction of the sample variation in y explained by the model.
Perfect collinearity	An exact linear relationship among the regressors; the model cannot be estimated.
Multicollinearity	High but not perfect correlation among regressors. Raises variances but causes no bias.
Zero conditional mean (MLR.4)	$E(u \mid x_1, \ldots, x_k) = 0$. The key assumption for unbiasedness; cannot be verified directly.
Homoskedasticity (MLR.5)	$\mathrm{Var}(u \mid x_1, \ldots, x_k) = \sigma^2$. Constant error variance.
Heteroskedasticity	Failure of MLR.5. Does not bias OLS but invalidates the usual standard errors.
Unbiasedness	$E(\hat{\beta}_j) = \beta_j$. A property of the procedure across repeated samples, not of any one estimate.
Omitted variable bias	$\mathrm{Bias}(\tilde{\beta}_1) = \beta_2 \delta_1$, arising when a relevant variable correlated with a regressor is left out.
Gauss-Markov theorem	Under MLR.1–MLR.5, OLS is the best linear unbiased estimator (BLUE).
Elasticity	The percentage change in y from a 1% change in x; the slope in a log-log model.
Semi-elasticity	The percentage change in y from a one-unit change in x; $100 \cdot \beta_1$ in a log-level model.

Review questions

Use these to check your understanding before the seminar.

1. Explain the difference between the error term u and the residual $\hat{u}$. Which one can you compute from data, and why does the distinction matter?
2. Galton found that tall parents have shorter children on average. Explain why this does not imply that human heights are converging over generations.
3. In the regression wage = −0.90 + 0.54 educ, the intercept is negative. Is this a sign that something is wrong with the estimation? Explain.

4. State which of these models OLS can estimate and which it cannot, and explain why: (a) $y = \beta_0 + \beta_1 x^2 + u$; (b) $y = \beta_0 + \beta_1^2 x + u$; (c) $y = \beta_0 + \beta_1 \log(x) + u$; (d) $\log(y) = \beta_0 + \beta_1 x + u$.
5. A researcher includes expendA, expendB and totexpend in a vote-share regression and EViews returns an error. Explain what has gone wrong in terms of MLR.3, and describe the ceteris paribus experiment that has become impossible.
6. Suppose the true model is wage = $\beta_0 + \beta_1$educ + $\beta_2$ability + u and you omit ability. Determine the sign of the bias in $\tilde{\beta}_1$ and explain each of the two signs you used.
7. Now suppose instead that the omitted variable were negatively correlated with education but still raised wages. What would the direction of the bias be?
8. Explain, using the variance formula, why adding a regressor that is highly correlated with x_1 makes $\hat{\beta}_1$ less precise – even though it introduces no bias.
9. MLR.4 fails and MLR.5 fails. For each case, say whether OLS is still unbiased, whether it is still BLUE, and whether the usual standard errors can be trusted.
10. A colleague reports $R^2 = 0.09$ and concludes the regression is worthless. Give two reasons this conclusion may be wrong.
11. If salary is measured in euros rather than thousands of euros, what happens to the slope, the intercept, and R^2? Justify your answer for each.
12. In log(wage) = 0.58 + 0.083 educ, interpret the slope. Explain why the dollar return to a year of education is not constant even though the percentage return is.
13. Why should you almost always include an intercept, even when theory suggests it is zero? Frame your answer as a comparison of what you risk in each case.

References and further reading

Wooldridge, J. M. Introductory Econometrics: A Modern Approach – Chapters 2 and 3 cover the material of this lecture.

Asteriou, D., Hall, S.G. Applied Econometrics – Chapters 3 and 4.